

13. Analysis D: minimum detectable size at 50%, 10%, and 5%, with a predicted 0% point

D₅₀, D₁₀, and D₅ come from a psychometric curve fitted to the detection trials, after correcting for false alarms. The curve is fitted in $\ln D_{\text{px}}$, pooled over the (feature, station) pairs of the Z sweep; all three minimums are read from the best of three fitted shapes by deviance, with the range across the shapes as the model uncertainty of each. D₅ is the lowest measured point; the extended trials at the far stations are sized for it. The 0% point is not measured: a smooth curve never reaches zero, and no finite number of trials can prove a probability is zero. D₀ is therefore reported as an extrapolation of the fitted curves to the 1% level (DETECTION_ZERO_PREDICTION_LEVEL), flagged as a prediction, and cross-checked against a model with a hard floor and, for cutouts, against the geometric limit. The prior expectation for the fitted D₅₀ and the predicted D₀ is the laser-pencil model of Section 3.2, about 7 px; the analysis reports where the data land against it.

1. Independence check. For each feature, compute the lag-1 autocorrelation of its detected/missed sequence in acquisition order. It should lie within $\pm 2/\sqrt{n}$. Also check that outcomes at neighboring features in the same frame are uncorrelated. Correlation at either level means the offsets were too small or the features too close; widen the 8 px span of the random offsets (PHASE_JITTER_SPAN_PX) or the 30 px feature isolation (FEATURE_ISOLATION_PX) and repeat.
2. Detection rule. In the feature window (radius $D_{\text{px}}/2$ plus a 3 px margin, DETECTION_WINDOW_MARGIN_PX), compute the deviation from the reference plane. For a disk, $\Delta = Z_{\text{back}} - Z$ (toward the sensor). For a cutout, $\Delta = Z - Z_{\text{front}}$ (away from the sensor). Candidates are valid reads with $\Delta > \tau$. The feature counts as detected if at least 2 connected candidates exist (DETECTION_MIN_CONNECTED_PX). That is the analysis's threshold for calling a trial detected, deliberately permissive, so that the sensor's own floor comes out of the data and is not set by the rule. No-reads do not count in the primary rule. A second, "no-read-inclusive" rule treats no-reads as candidates for cutouts, since a downstream reader might take a hole of no-reads to be a hole. Report both, because the gap between them is a direct measure of boundary bias (Section 14).
3. Threshold. For each Z, configuration, and window size, apply the rule to the blank sites and set τ so that the false-alarm fraction equals the 1% target (DETECTION_FALSE_ALARM_TARGET). Pool the blank windows of a station over its blank sites; at a main station that is at most 3 sites $\times$ 60 trials = 180 windows, for the smallest window size, and fewer for the larger ones, because each blank site is sized to one feature and hosts only the windows no larger than its own; the 99% quantile then falls among the largest few values and is poorly determined. Where a station has

fewer than 300 blank windows (DETECTION_MIN_BLANK_WINDOWS, three per false alarm at the target), divide the window statistic by $\sigma_{\text{tot}}(Z)$ from Section 10 and pool the neighboring stations until the count is reached; the extended stations reach it on their own. Then record the measured false-alarm rate γ with its Clopper–Pearson interval beside τ , so that the control achieved is visible.

4. Psychometric fit. Model the raw detection probability as below, with $x = \ln D_{\text{px}}$, pooled over the features and stations. F is the logistic function, γ is fixed from the blank sites at each station, and the lapse rate λ is free but bounded between 0 and 0.05 (DETECTION_LAPSE_RATE_MAX). Fit α, β, λ by maximum binomial likelihood. The false-alarm-corrected probability is $P^*(D) = F$.

$$\psi(D) = \gamma + (1 - \gamma - \lambda) F\left(\frac{\ln D - \alpha}{\beta}\right), \quad D_{50} = e^{\alpha}, \quad D_{10} = e^{\alpha - \beta \ln 9}$$

5. Model dependence. D_{10} sits in the tail, so it depends on the curve shape more than D_{50} does. Fit three shapes for F : logistic, cumulative normal, and Weibull in $\ln D$. Report D_{50} , D_{10} , and D_5 from the best fit by deviance, and the range across the three as the model uncertainty of each. As a check that needs no model, fit a monotone curve to the corrected proportions with isotonic regression and interpolate the 0.1 and 0.5 crossings.
6. D_5 , the lowest measured point. At each level, compute the two-sided Clopper–Pearson interval of the raw detection rate at 95% confidence (CONFIDENCE_LEVEL) and correct it for false alarms. Read D_5 from the fitted curve at the corrected probability of 5% (DETECTION_LOW_PROBABILITY), with its bootstrap interval, and report beside it the measured corrected rates and intervals of the levels nearest to it, so that the reader sees the point is measured, not extrapolated.
7. D_0 , predicted (flagged). Extrapolate each fitted shape (logistic, cumulative normal, and Weibull in $\ln D_{\text{px}}$) below D_5 to the 1% level (DETECTION_ZERO_PREDICTION_LEVEL) and report the size where it crosses; give the range across the three shapes as the model uncertainty, and a profile-likelihood interval from the threshold model, a Weibull with a location parameter ($P^* = 0$ for $D \leq D_0$). Label every reported D_0 as a prediction from the fitted curve, never as a measurement; the measured curve ends at D_5 .
8. D_0 , geometric limit (cutouts). Find the diameter at which A_{geo} from Section 12, Step 7 reaches zero, meaning no part of the back plate is seen by both IR cameras and lit by the projector, the views a depth read needs. Below it, the primary rule cannot detect a cutout on any sensor of this baseline. A predicted D_0 well above the geometric limit means the sensor's processing, not geometry, sets the floor.

9. Confidence intervals. Bootstrap over poses with 2000 resamples (BOOTSTRAP_RESAMPLES), re-fitting γ , τ , and the curve each time. This gives intervals for D_50, D_10, D_5, and the predicted D_0.
10. Units, scaling, and Z dependence. Report each minimum in three forms: D_px, $\theta = D_{\text{px}} / f_x$ in mrad, and D in mm at each station. The Z sweep makes the noise covariate explicit: fit a logistic regression on $\ln D_{\text{px}}$ and $\ln \sigma_{\text{tot}}(Z)$ from Section 10 together. A negligible σ_{tot} term means the minimum is set by pixel or projector-dot sampling; a significant one means it is set by noise. Apply the scaling test of Section 12 here too: the detection curves of neighboring features must agree over their shared D_px range, and a disagreement is reported with its σ_{tot} attribution, one row per pair of neighboring features, kind, gap, field position, and rule in D_overlap_test.csv.
11. Outputs. D_detect_summary.csv with one row per configuration and station, and D_pooled_summary.csv with one pooled row per configuration; D_overlap_test.csv; D_detect_details.json. Columns: γ , τ , D_50, D_10, D_5, the predicted D_0 with its model range, geometric D_0 (cutouts), and confidence intervals, each in px, mrad, and mm, for both detection rules. Figures (PNG and SVG): psychometric curves in D_px with binomial error bars, D_5, D_10, and D_50 marked and the predicted D_0 drawn as a dashed extrapolation, one trace per feature; θ against Z for each minimum. D_50 and D_10 in px go in forward_model_parameters.json for the Tier-A simulator.